

№6

2026-09-04

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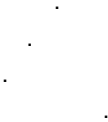
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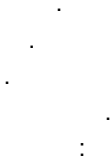
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```
1 # Import the necessary libraries
2 import pandas as pd
3 import numpy as np
4 import matplotlib.pyplot as plt
5 import seaborn as sns
```

```
6 # Load the data
7 data = pd.read_csv('data.csv')
8 # Display the first 5 rows of the data
9 data.head()
```

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$$\begin{aligned}
 & \text{; - R (} \quad \quad \quad \text{) -} \\
 & \quad \quad \quad \text{: - S (} \quad \quad \quad \text{) -} \quad \quad \quad \text{: - I (} \quad \quad \quad \text{) -} \\
 & \quad \quad \quad \text{/} \quad \quad \quad \text{) -} \quad \quad \quad \text{.} \\
 & \text{: - infection: } S + I \rightarrow I + I \text{ (} \quad \quad \quad \beta \text{); - recovery: } I \rightarrow R \text{ (} \quad \quad \quad \gamma \text{)}
 \end{aligned}$$

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DrWatson julia (. 1).

```
julia> Pkg.add("DrWatson")
    Updating registry at `~/ .julia/registries/General.toml`
└─ Error: Some registries failed to update:
    └─ /home/mwakutaipa/.julia/registries/General.toml - failed to d
ownload from https://pkg.julialang.org/registry/23338594-aafe-5451-b9
3e-139f81909106/31c1bca344f440e166a96c5b4dca8b6c99b1b120. Exception:
RequestError: HTTP/2 200 (Recv failure: Connection reset by peer) whi
le requesting https://pkg.julialang.org/registry/23338594-aafe-5451-b
93e-139f81909106/31c1bca344f440e166a96c5b4dca8b6c99b1b120
└─ @ Pkg.Registry ~/ .julia/juliaup/julia-1.10.10+0.x64.linux.gnu/share
/julia/stdlib/v1.10/Pkg/src/Registry/Registry.jl:528
    Resolving package versions...
    No Changes to `~/ .julia/environments/v1.10/Project.toml`
    No Changes to `~/ .julia/environments/v1.10/Manifest.toml`

julia> using DrWatson

julia> initialize_project("project"; authors="Mildred", git=false)
```


(. 2)

```
julia> cd("project")

julia> Pkg.add("AlgebraicPetri")
  Resolving package versions...
  Installed XML2_jll ————— v2.15.1+0
  Installed NameResolution ——— v0.1.5
  Installed LabelledArrays ——— v1.19.0
  Installed GeneralizedGenerated — v0.3.3
  Progress [=====> ] 4/22
```

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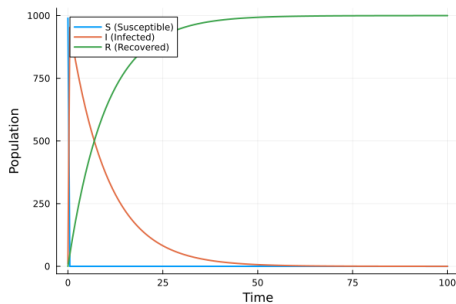
beta gamma,

:

(. 3)

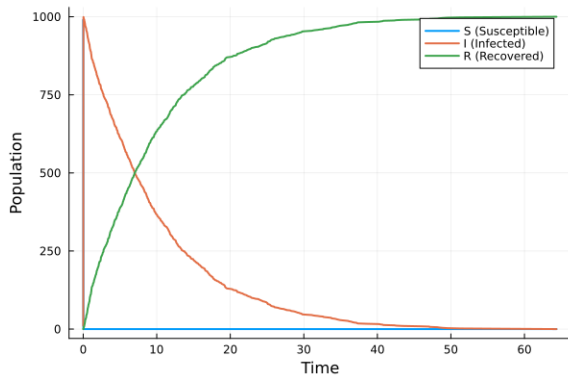
(. 4).

CSV



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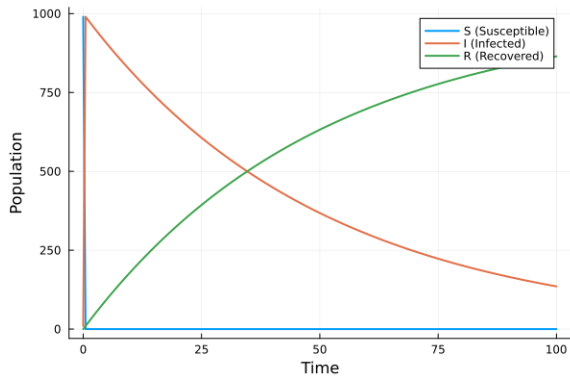
SIR

Детерминированная симуляция

```
] : for beta in β
    for gamma in γ
        df_det = simulate_deterministic(net, u0, (0.0, tmax), saveat = 0.5, rates = [beta, gamma])
        CSV.write(datadir("sir_det.csv $beta, $gamma"), df_det)
        println("simulation for $beta, $gamma")
    end
end

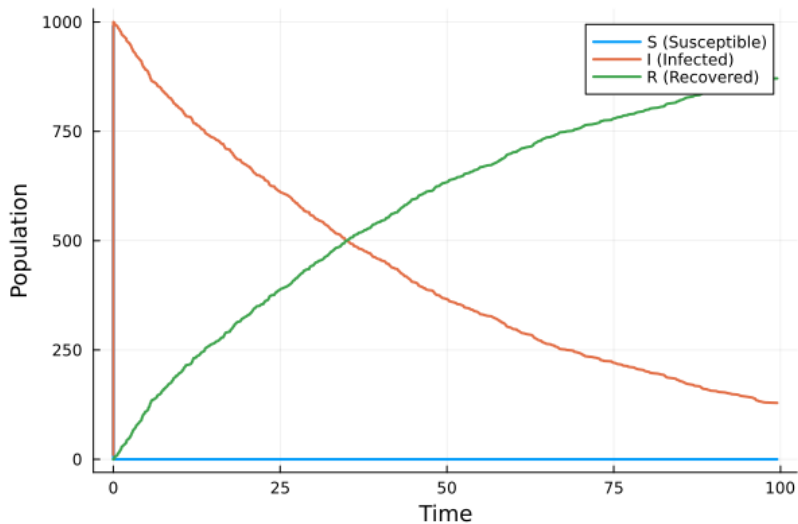
simulation for 0.9, 0.6
simulation for 0.9, 0.3
simulation for 0.9, 0.02
simulation for 1.0, 0.6
simulation for 1.0, 0.3
simulation for 1.0, 0.02
simulation for 0.75, 0.6
simulation for 0.75, 0.3
simulation for 0.75, 0.02
```

5:



6:

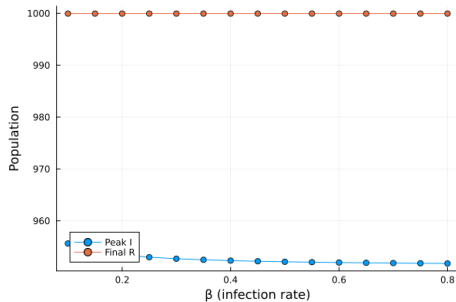
SIR



β .

(. 8)

CSV

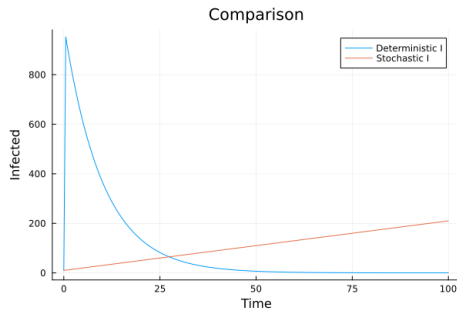


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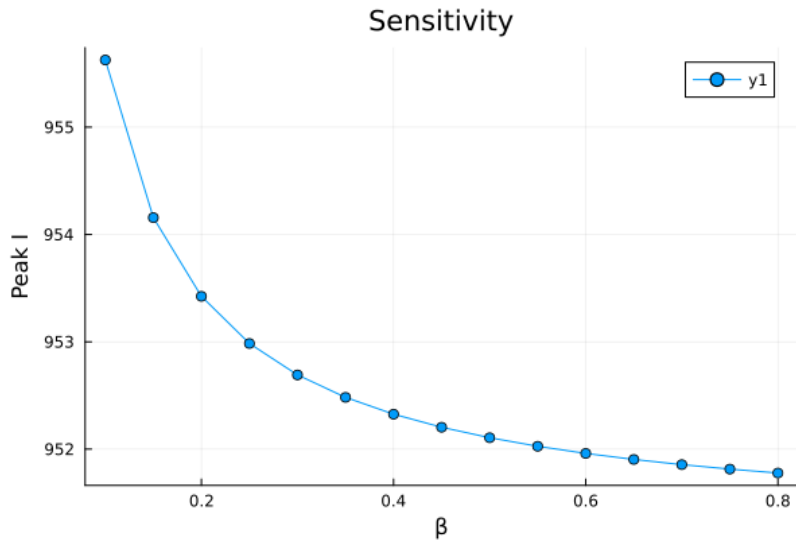
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